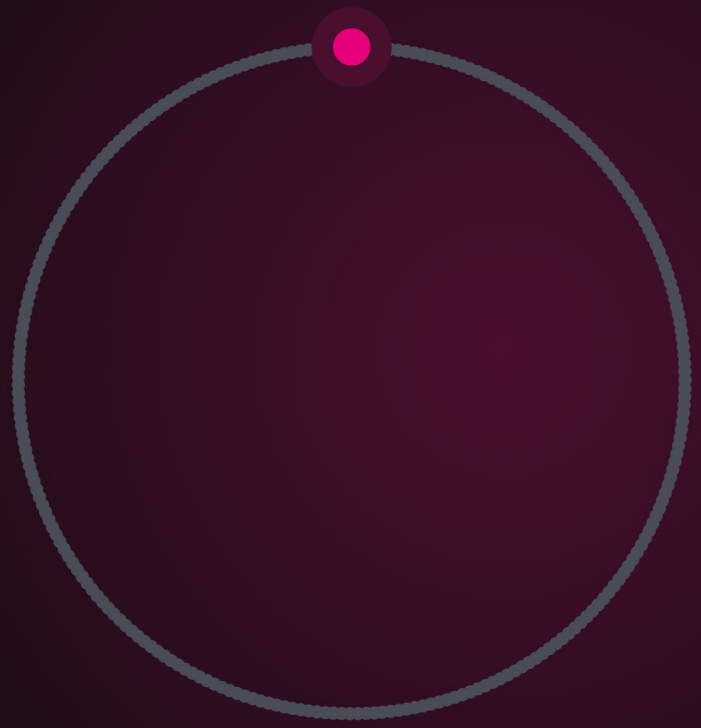
POLKADOT ASSEMBLE · MUMBAI Sat 7 Nov 2026 · Main stage

The Backend You Don’t Run

Private messaging, storage and personhood on Polkadot



Dhiraj Parity Technologies

THE PROBLEM

Count the servers

Each box is someone else's server, with a recent incident.

6 operators in a typical 2026 hackathon dApp

Frontend host

Vercel, DNS, eth.limo

eth.limo DNS hijack, 18 Apr 2026

RPC provider

Infura, Alchemy

Infura cut off Venezuela, 3 Mar 2022

IPFS + pinning

Pinata, the ipfs.io gateway

ipfs.io gateway sunset, 21 Sep 2026

Chat backend

XMTP and others

XMTP's hosted network closes, early 2027

WebRTC signaling

Usually your own server

One more box to run and pay for

Bot filter / KYC

World ID, KYC vendors

World ID paused in India since Dec 2023

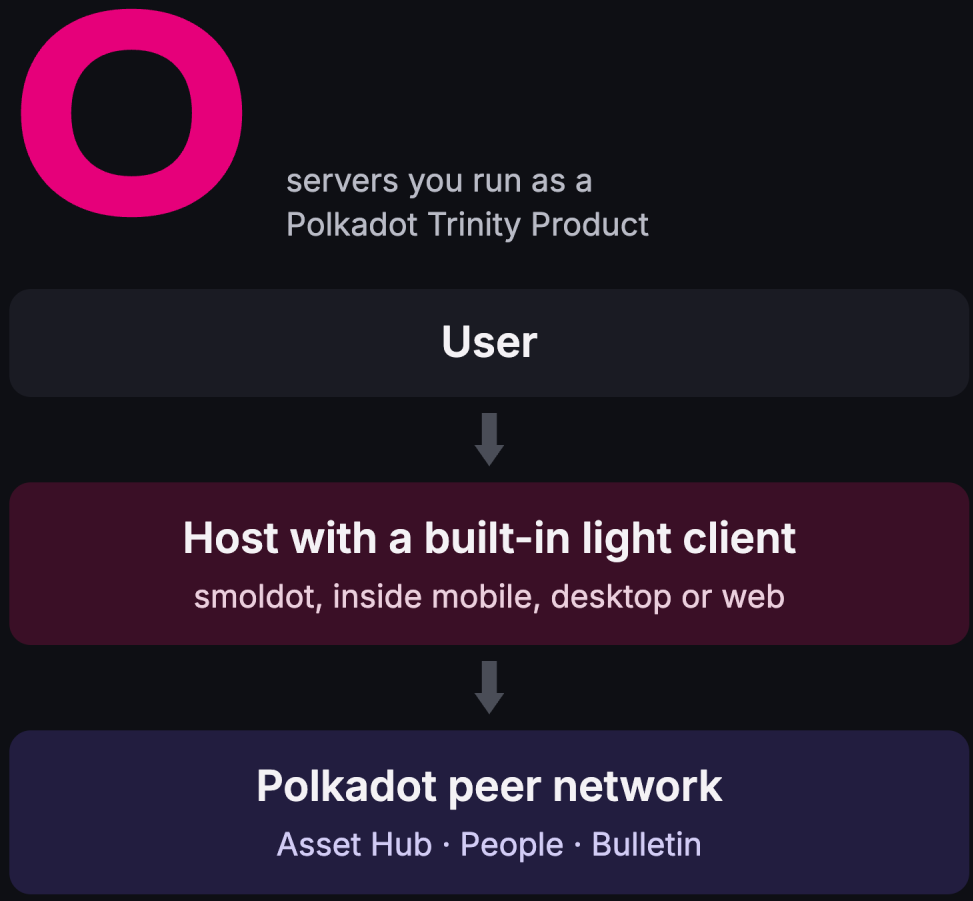
Ethereum has the building blocks. The gap is the defaults.

THE IDEA

What if the backend were the network?

Your dApp's backend can be the network itself: private messaging, storage and personhood, through a light client.

...and you run no servers to do it.



Optional helpers exist (call relays, the dot.li web gateway, onboarding), run by the Polkadot Community Foundation. None is trusted for correctness.

THE STACK

The shape of the stack

PRODUCT

A sandboxed single-page app at name.dot

It can't hold keys, make arbitrary network requests or talk to chains directly.

↓ TrUAPI frames, SCALE-encoded, over a MessagePort or postMessage channel

HOST · MOBILE, DESKTOP OR WEB (DOT.LI)

Keys and signing

Held only by the mobile host

Shared Rust core

The TrUAPI runtime: WASM on the web, UniFFI on iOS and Android

→

smoldot light client

The only route to the chains

↓ Peer to peer: every answer is checked, not believed

Asset Hub

`para 1000`

Contracts, DotNS, payments

People

`para 1004`

Personhood, allowances, statement-store gossip

Bulletin

`para 1010`

Content-addressed data, Bitswap

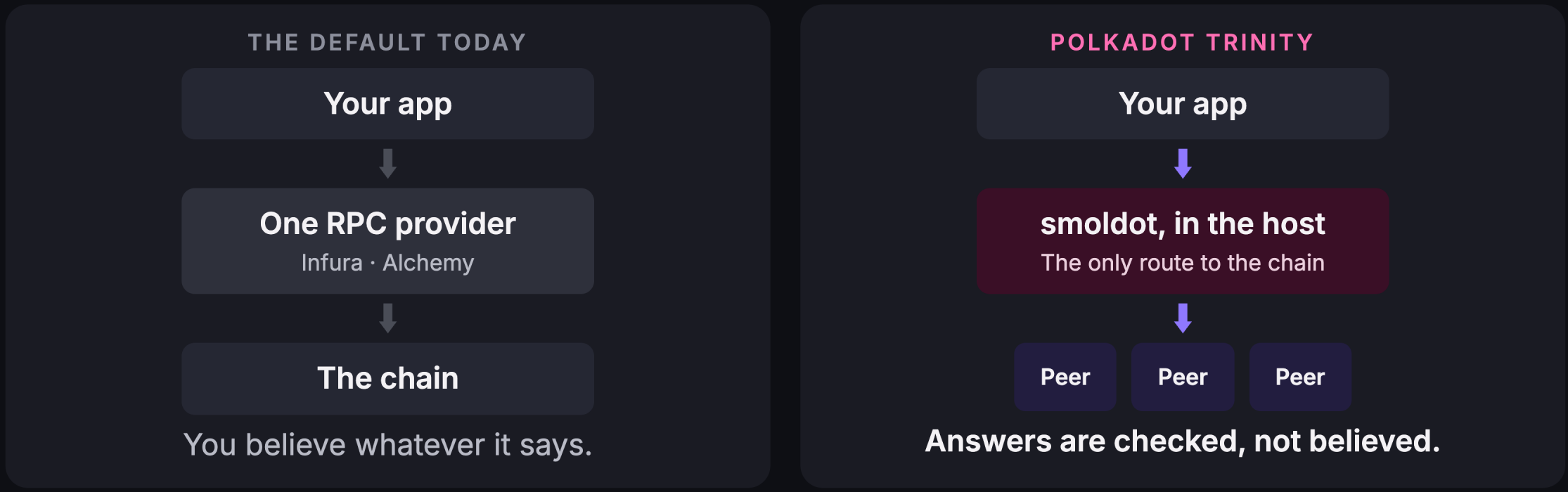
Relay chain

`Polkadot`

Security and finality

THE STACK

The light client is the connection



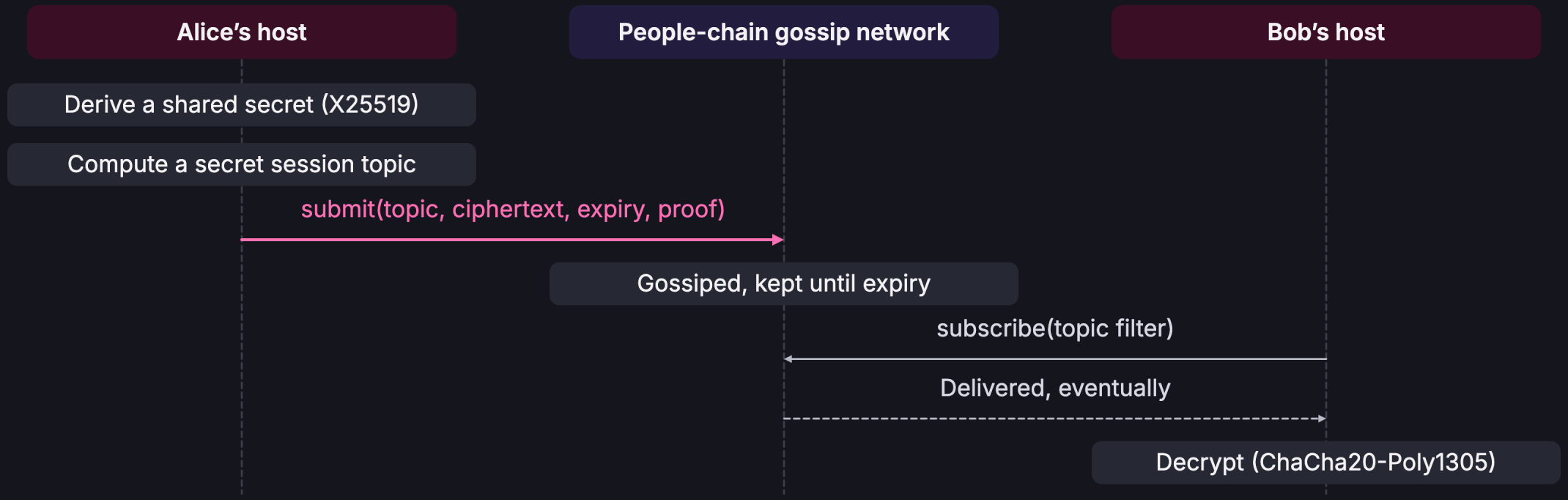
No major Ethereum wallet ships a light client by default. Defaults decide.

BUILDING BLOCK 1 · CELERITY (STATEMENT STORE)

Private messages, no server

Chain layer live on People Polkadot · 10 Sep 2026

Apps on testnet



Store-and-forward pub/sub

≤ 4 topics · ≤ 1 MiB · an expiry

"Closer to UDP than TCP"

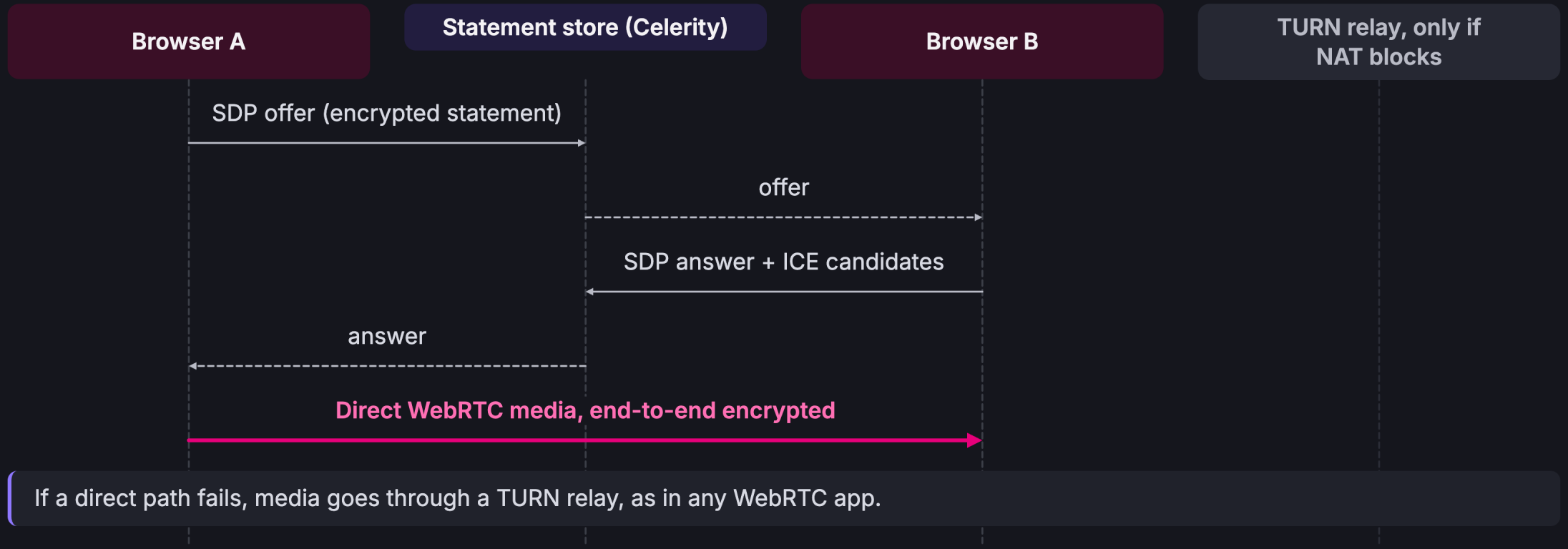
The store doesn't encrypt. Apps do.

...and you run no servers to do it.

BUILDING BLOCK 1 · CELERITY (STATEMENT STORE)

Calls with no signaling server

The same pattern carries presence, private notifications and coordination.



If a direct path fails, media goes through a TURN relay, as in any WebRTC app.

BUILDING BLOCK 2 · LEVITY (BULLETIN CHAIN)

Storage that expires, on purpose

Para 1010 live · storage since 1 Sep 2026

App publishing on testnet

Authorized
An authorizer grants an allowance

→

Stored
store(data) returns a CID

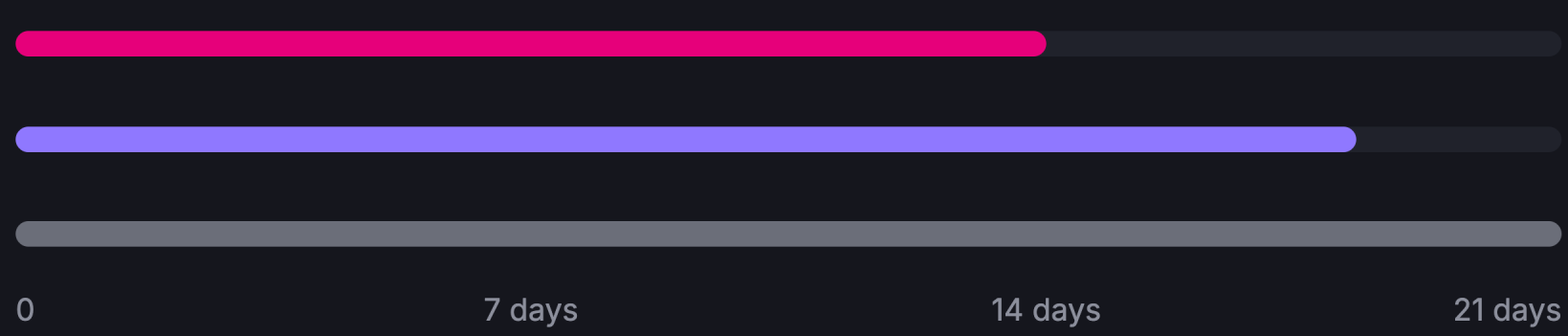
→

Retained
201,600 blocks ≈ 14 days

→

Expired
If nobody renews it

Renew keeps data retained, up to a hard per-account cap.



CID + Bitswap through the light client, no gateway. Whole apps live on .dot names.

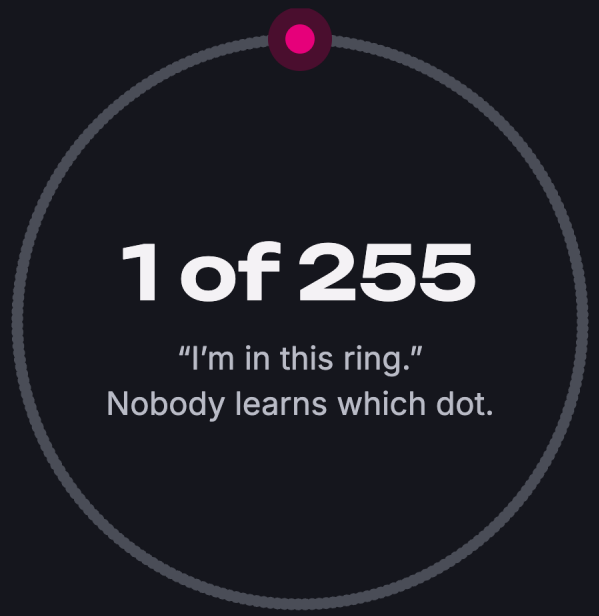
...and you run no servers to do it.

BUILDING BLOCK 3 · HUMANITY (PROOF OF PERSONHOOD)

Prove you're a person, not which person

Pallets live on People Polkadot

Recognition game on testnet



A weekly game in groups of six. No iris scan, no Aadhaar.

SAME PERSON · THREE APPS · THREE ALIASES

Chat app	alias A
Marketplace	alias B
Community poll	alias C

~100 ms
to prove

~5 ms
to verify

THE TWIST

Personhood is the rate limiter

Messaging and storage quotas go to people, not wallets.

WHO	STATEMENTS	DATA
Lite person	15	50 KiB
Full person	30	100 KiB
Alias slot	2	500 KiB
Notification	1	10 KiB

Alias slots go to an unlinkable account: 20 a day for persons, 10 for lite persons.
Bulletin storage via People: 1 MiB per claim, up to 100 claims per 2 weeks.

Spam control by person, not by purse.

WAKU	by stake (RLN)
XMTP	by fees
TRINITY	**by person**

PUT TOGETHER · A SKETCH

A bot-free community app

An app for events like this one, built only from the three blocks.

HUMANITY

Every attendee is a real person, under an alias

CELERITY

Private chat, presence and notifications

LEVITY

The schedule, the media and the site itself

LIGHT CLIENT

Every read checked, no RPC provider

ONE APP

assemble.dot

Verified humans, private chat, the schedule on Bulletin, a site at a .dot name

No backend.
No infra team.
No server bill.

...and you run no servers to do it.

PUT TOGETHER, HONESTLY · LIVE HAS LAYERS

Where it stands today

Live on Polkadot

Chain layer

- Bulletin chain, para 1010 · 1 Sep 2026
- Statement store + personhood pallets on People · 10 Sep 2026
- DotNS contracts, signed stack · 1 Nov 2026
- Lite-person onboarding (Ref #1958) · 2 Nov 2026

Build on testnet

Developer kit

- Paseo Next v2 and previewnet
- TrUAPI v0.29 · weekly releases, a prototype
- Personhood game, Proof-of-Ink, Mob Rule
- Publishing apps to .dot names

Users

No public mainnet numbers yet, so none here. The strongest builder signal: about 190 developers deployed apps and registered 272 .dot domains at Web3 Summit (testnet, 20 Jul 2026).

FOR ETHEREUM DEVELOPERS · AS OF NOV 2026

How it compares

Need	Ethereum world today	Polkadot Trinity	Where Ethereum is stronger
Node access	Infura or Alchemy by default; Helios is opt-in	smoldot inside the host, on by default	Helios is mature: the gap is the default
Messaging	XMTP, Waku, Session	Gossiped statement store; apps encrypt; personhood as anti-spam	XMTP's adoption and MLS; Waku's peer-reviewed RLN
Storage	IPFS + pinning, Arweave, blobs, ENS + eth.limo	CID + Bitswap through the light client; 14 days, renewable	Arweave is permanent: about $49/GiB once
Personhood	World ID, Self, Anon Aadhaar	A social game: no biometrics, no state ID	Scale: World has about 18M Orb-verified

WHERE IT'S GENUINELY DIFFERENT

Light client by default · One stack, not four vendors · Personhood without biometrics or ID · Spam control by person

START · TESTNETS ONLY

Start today

From zero to a .dot app on Paseo Next v2, in three steps.

shell

```
# 1 · Run your Product with a local signing host
# installer: github.com/paritytech/trinity-user-agents
$ curl -fsSL …/scripts/truapi-host-installer.sh | bash
$ truapi-host dev -- yarn dev:paseo

# 2 · Add the SDK
$ pnpm add @parity/product-sdk

# 3 · Ship a static app to a .dot name
$ npm install -g bulletin-deploy
$ bulletin-deploy --env paseo-next-v2 ./out your-name.paseo
# → https://your-name.paseo.li
```

Quick start docs
docs.polkadot.com/apps/quick-start

Pin your versions: TrUAPI is 0.x and ships weekly.

Count your servers.
Then delete them.

github.com/paritytech

docs.polkadot.com/apps

Thank you. Questions?